AI use cases for telecom operations and growth

Fifteen ways to improve service, efficiency and revenue, backed by a live Fibre and NOC Assurance platform.

Customer value and revenue growth

Stronger retention, faster B2B proposals and new paid services.

Revenue from existing assets

Faster data-centre activation and higher sales from spare fibre and capacity.

Network reliability and delivery

Pre-cut warning, street-accurate fault pinpointing and shorter outages.

Fifteen opportunities across three business priorities

Customer value and B2B growth

01 Spam and fraud protection
Lower losses and paid protection

02 Voice analytics
Better service and sales follow-up

03 Next-best offer and retention
Better offers and lower churn

04 AI on business plans
New subscription revenue

11–13 B2B upsell, proposals and sentiment
Verified RFP route bids and account health

Revenue from existing assets

05 Unused capacity
Sell spare dark fibre, wavelengths, voice and SMS

06 Data-centre interconnect
Verified diverse paths and faster activation

07 Roaming analytics
Better margins and roaming usage

14 PUE and energy management
Lower data-centre cooling waste

15 AI-compute capacity
More useful compute from GPU clusters

Network reliability and delivery

08 Fibre-cut assurance
Dig warnings, slack-adjusted GPS and splice checks

09 Incident resolution
Alarm correlation and false-dispatch filtering

10 Optical planning
Zero-shared-duct routes and less overbuild

Live platform demonstration

Use cases 06, 08, 09, 10 and 12 are running live and illustrated in the product views that follow.

Customer value and revenue growth

01 Spam and fraud protection

Lower fraud losses and a protection service to sell

Risk alerts and prioritised blocking or investigation actions for suspicious calls and messages.

Measures: Fraud losses and paid subscriptions

02 Voice analytics

Better service and more relevant sales follow-up

Recurring issues, sales opportunities and service quality findings from consented call recordings.

Measures: Call review time and issue resolution

03 Next-best offer and retention

More customers retained at a lower discount cost

Offers and retention actions based on each customer’s usage, service experience and complaints.

Measures: Retention rate and discount cost

04 AI assistants on business plans

New subscription revenue from business customers

Paid assistants for enquiries, appointment booking and service requests, connected to customer tools.

Measures: Paid adoption and revenue per account

Revenue from existing assets

05 Unused capacity monetisation

More revenue from dark fibre, wavelengths, ports, voice and SMS capacity

Identifies unlit fibre cores, spare 100G/400G optical channels and off-peak capacity matched to wholesale and enterprise demand, pricing and contract limits.

Data: GIS cable inventory, port occupancy, rates and contracts Measures: Extra revenue, margin and capacity sold

06 Data-centre interconnect

Faster customer activation and earlier billing

Instant diverse-path checks between data centres (confirming zero shared ducts or bridges), glass latency calculation, and pre-staged provisioning workflows.

Data: Optical ports, GIS duct routes, latency targets and orders

Measure: Quote turnaround and order-to-activation time

07 Roaming analytics

Better roaming margins and more customer usage

Margin leakage reports, partner rate comparisons and targeted travel offers for subscribers who do not currently use roaming.

Data: Roaming usage, partner rates and billing

Measure: Roaming margin and active users

Network reliability and delivery

08

Fibre-cut assurance

Early warning and street-accurate repair

WHAT YOU GET

Pre-cut digging vibration alerts, binary OTDR trace analysis subtracting coiled manhole slack (avoiding 150–300m dig errors), subsea anchor watch and splice sign-off.

Data: OTDR traces (.sor), GIS ducts and slack, optical SOP and AIS

Measures: Prevented cuts, restoration time and site visits

09

Incident resolution

Shorter outages and fewer wasted dispatches

WHAT YOU GET

Groups optical, IP and 5G alarm storms into one root cause, rules out site power loss, microwave rain fade and planned roadworks, and stages human-approved reroutes.

Data: Multi-vendor alarms, change tickets, GIS and SLA inventory

Measures: Triage minutes, avoided truck rolls and SLA credits saved

10

Optical planning

Verified route diversity and less overbuild

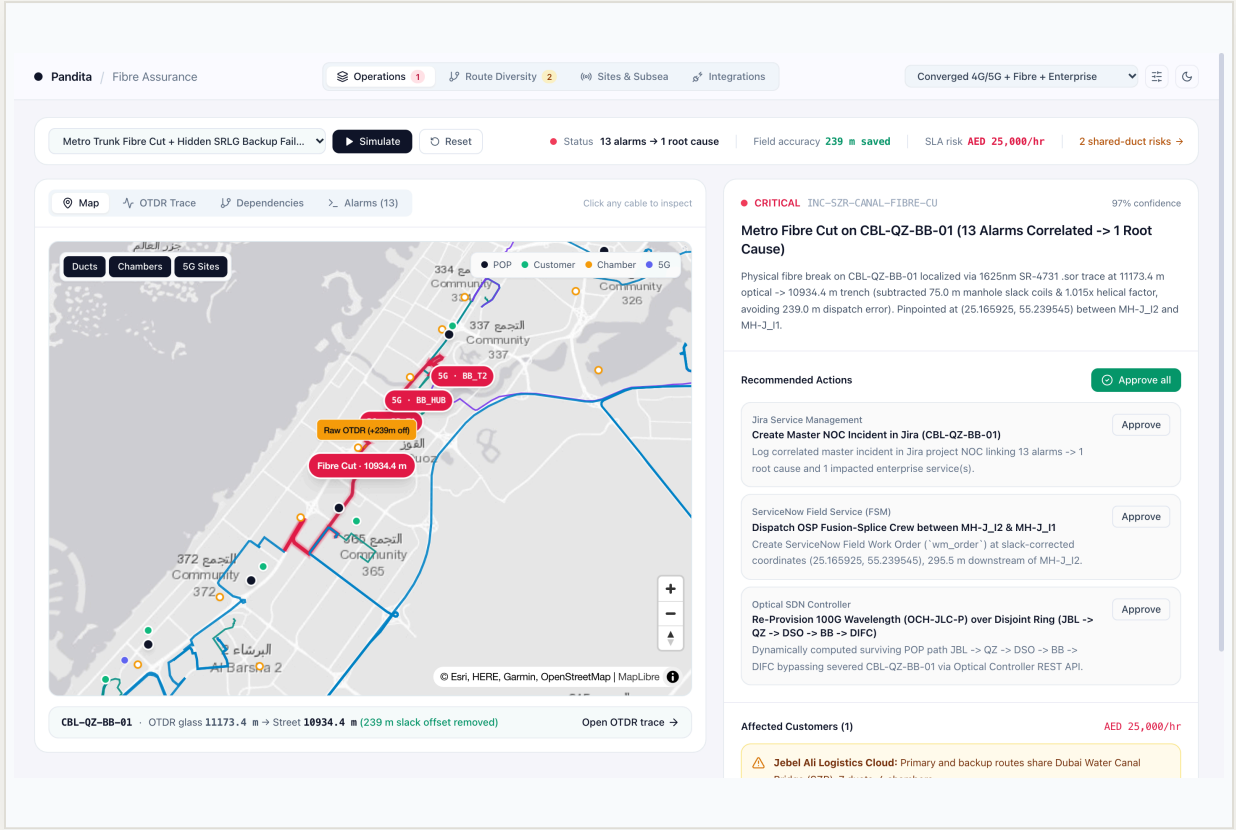
WHAT YOU GET

Primary and backup optical route designs guaranteed free of shared chambers, ducts or bridges, complete with optical loss budget and glass latency checks on existing plant.

Data: GIS ducts and cables, DWDM spans, fibre counts and orders

Measures: Design time (seconds vs. days) and build cost saved

Single-screen fibre-cut pinpointing and human-gated recovery



13 alarms → 1 physical root cause

Collapses optical LOS, IP/BGP drops and 5G cell alarms onto trunk cable CBL-QZ-BB-01 within seconds, ranking AED 25,000/hr SLA exposure first.

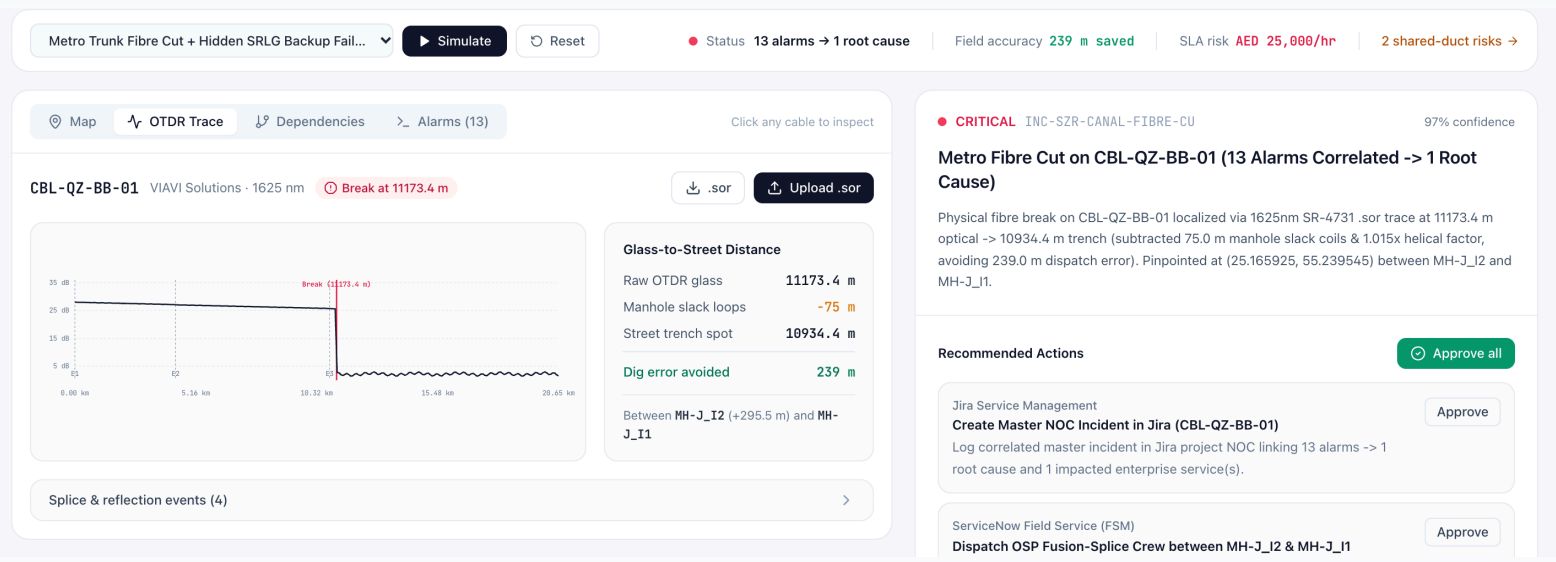
239 m dig error avoided on the map

Plots both the raw optical distance and the slack-corrected street trench point between chambers MH-J_I2 and MH-J_I1 so crews dig once.

1-click engineer-approved execution

Stages the Jira master ticket, ServiceNow splice dispatch and 100G disjoint-ring optical reroute behind explicit NOC approval gates.

Binary OTDR (.sor) trace parsing with manhole-slack subtraction



Native SR-4731 (.sor) parsing

Reads 1625 nm reflectometry waveforms directly from VIAVI and EXFO test heads without waiting for manual PDF exports.

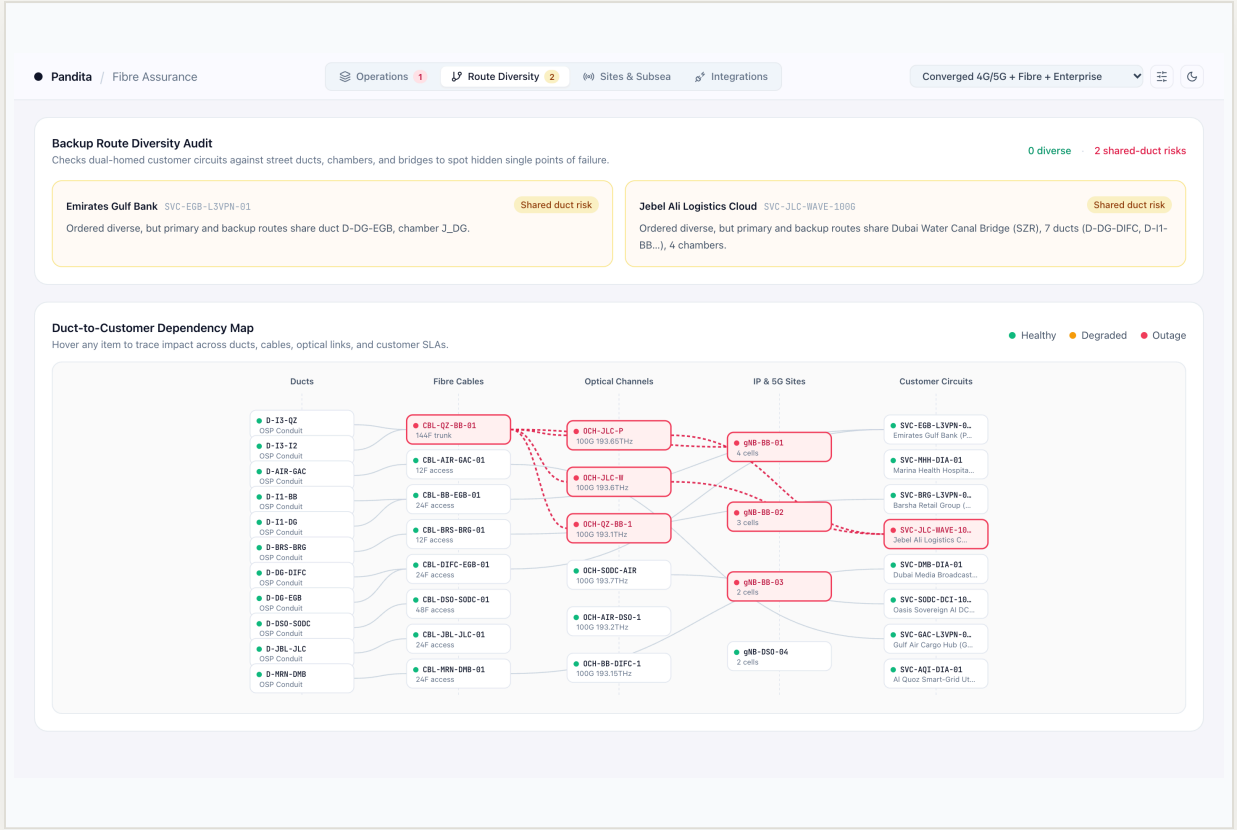
Glass-to-street conversion

Subtracts 75 m of coiled manhole slack and 1.015x cable twist to turn 11,173.4 m of glass into 10,934.4 m on the street.

Remote splice quality sign-off

Verifies post-repair fusion splice loss (≤0.15 dB) before field technicians seal the chamber and close the work order.

Shared-duct diversity audit and duct-to-customer impact graph



Exposes hidden **shared-duct failures**

Flags dual-homed circuits whose primary and backup paths quietly cross the same bridge, chamber or street conduit before a cut takes both down.

5-layer **duct-to-SLA visibility**

Maps every underground conduit through fibre trunks, 100G optical channels and 5G towers directly to affected enterprise circuits.

Seconds vs. days **for DCI & RFP bids**

Gives pre-sales and optical planners instant proof of diverse paths and glass latency for hyperscale, banking and wholesale quotes.

AI for Telco B2B Growth & Sales

11

Cross-sell and up-sell

More revenue from existing accounts

PROPOSED APPROACH

Ranks account opportunities from bandwidth usage, contract gaps and single-duct route exposure so B2B teams can pitch concrete resilience upgrades.

Data: CRM, usage, contracts, route diversity audits and catalogue

Measures: Offer conversion and revenue per account

12

Solution and proposal quality

Faster proposals with verified routes

PROPOSED APPROACH

Generates GIS-verified diverse route maps and optical latency sheets in seconds for enterprise RFPs, while flagging technical or SLA gaps before sign-off.

Data: RFPs, GIS duct inventory, SLA catalogue and past proposals

Measures: Proposal cycle time and rework

13

Customer sentiment

Earlier action on account concerns

PROPOSED APPROACH

Analyses consented calls, trouble tickets and SLA breach history to surface recurring service issues and renewal risks for account-team follow-up.

Data: Customer interactions, NOC tickets and surveys

Measures: Issue resolution time and renewal rate

AI for Data-Centre Efficiency & Capacity

14

PUE and energy management

Lower energy waste within operating limits

PROPOSED APPROACH

Combine facility and IT energy data with cooling telemetry. Detect unusual consumption and recommend cooling adjustments for operator approval.

Data: Power meters, cooling telemetry, IT load and weather

Measures: PUE, total kWh and thermal-limit breaches

Baseline comparisons account for IT load and ambient weather.

15

AI-compute capacity optimisation

More useful compute from available capacity

PROPOSED APPROACH

Forecast GPU demand and identify bottlenecks across power, cooling and compute. Recommend workload placement and capacity plans within service targets.

Data: GPU utilisation, job queues, rack power and cooling limits

Measures: GPU utilisation, queue time and SLA attainment

Placement decisions retain workload isolation and service targets.

Six capabilities ready across metro, national and subsea networks

01 Pre-cut digging warning

Spots mechanical digging vibration along buried ducts and switches traffic to a backup path before the cable snaps.

Fit: Metro & Dark Fibre (e.g., Constl)

02 Slack-adjusted street GPS

Subtracts coiled manhole slack from raw OTDR traces for exact dig coordinates, while ruling out site power loss and rain fade.

Fit: Fixed, 5G Backhaul & Metro Carriers

03 Pre-sales diverse route sheet

Checks two customer sites in seconds to confirm zero shared ducts or bridges, exporting a route and latency sheet for RFPs.

Fit: DCI, Hyperscale & Enterprise Sales

04 Subsea vessel & wet-plant watch

Overlays live ship positions (AIS) onto shallow cable corridors alongside repeater power and C-OTDR telemetry to catch anchor drag.

Fit: Subsea & Wholesale (e.g., PCCW Global)

05 Maintenance & roadwork match

Cross-checks alarms against active change windows and municipal roadwork permits within 500m to flag planned work immediately.

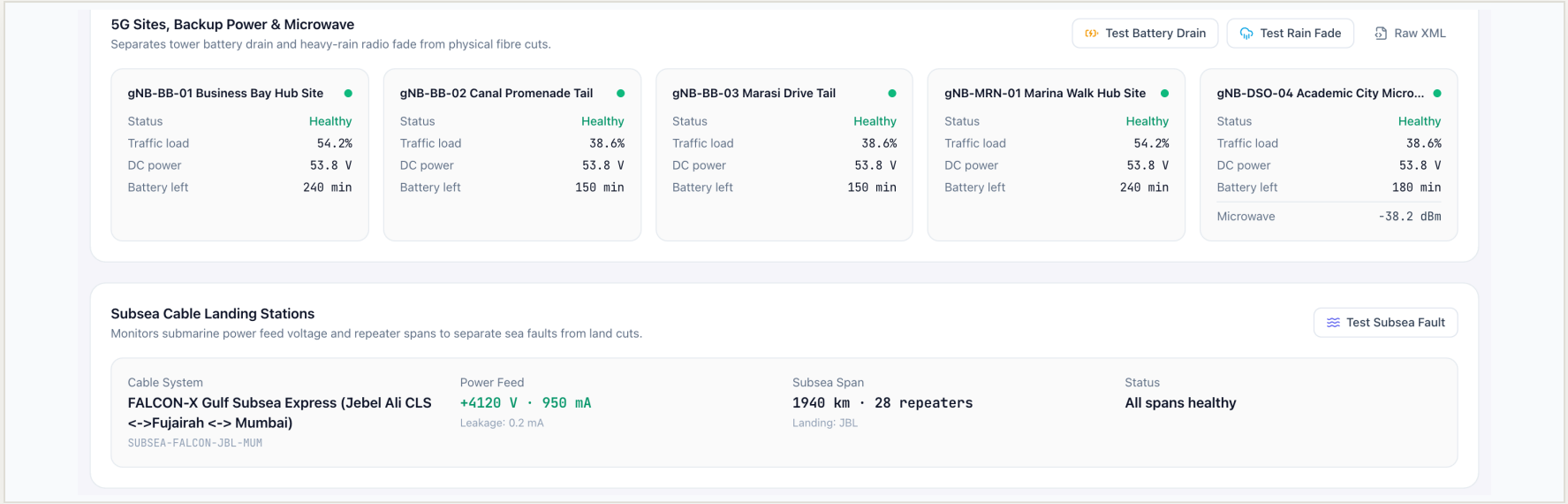
Fit: National Telcos (e.g., e&, du, Vi)

06 Field sign-off & customer RFO

Sends tube and core colour codes to crews, verifies splice loss (≤0.15 dB) before ticket closure, and drafts customer outage reports.

Fit: Field Operations & Enterprise SLAs

Separating power loss, rain fade and subsea faults from fibre cuts



Tower power & battery check

Reads 5G site DC voltage and battery reserve before rolling a fibre crew, routing commercial power blackouts to facilities instead.

Microwave rain-fade filter

Distinguishes heavy-rain E-band attenuation (-38.2 dBm) on tail sites from physical backhaul breaks to stop wasted dispatches.

Subsea landing & wet-plant watch

Tracks +4120 V power-feed telemetry and 28 repeater spans from the landing station to separate wet-plant faults from terrestrial cuts.

Grounded language models, vector RAG and signal analytics

01 Grounded LLMs for NOC & RFO synthesis

Anthropic Claude Sonnet 4, OpenAI GPT-4o, or on-prem Llama 3.3 / Qwen 2.5 (vLLM)

Drafts cited NOC incident briefs, step-by-step repair procedures, multilingual customer RFOs (EN/AR) and RFP answers strictly from verified JSON telemetry, with an offline fallback.

Guardrail: Zero free-form generation of GPS coordinates, circuit IDs or SLA figures

02 Vector embeddings & Runbook RAG

PostgreSQL pgvector + OpenAI text-embedding-3, local BGE, or 64-D telecom embeddings

Embeds live multi-vendor alarm symptoms and retrieves the closest approved carrier Methods of Procedure (MOPs) and historical resolved incident tickets via cosine similarity.

Corpus: Operator SOPs, vendor alarm dictionaries and historical MTTR post-mortems

03 Causal graph & fault-signature classifiers

NetworkX 6-layer dependency clustering + probabilistic root-cause scoring

Partitions 10,000-alarm bursts across ducts, DWDM, IP and 5G into independent fault clusters and scores competing hypotheses (97% fibre break vs. power loss, battery drain or rain fade).

Outcome: Sub-second storm collapse and automated false-dispatch suppression

04 Optical & vibration anomaly detection

NumPy signal & slope analytics over .sor waveforms, Pre-FEC BER and SOP streams

Detects mechanical digging vibration (polarization transients in rad/s), non-reflective splice pinches and Pre-FEC BER cliff slopes 15–40 minutes before complete span failure.

Outcome: Proactive hitless wavelength switchover before customers see packet loss

Deterministic core connected across twelve carrier systems

Spatial & optical physics engine

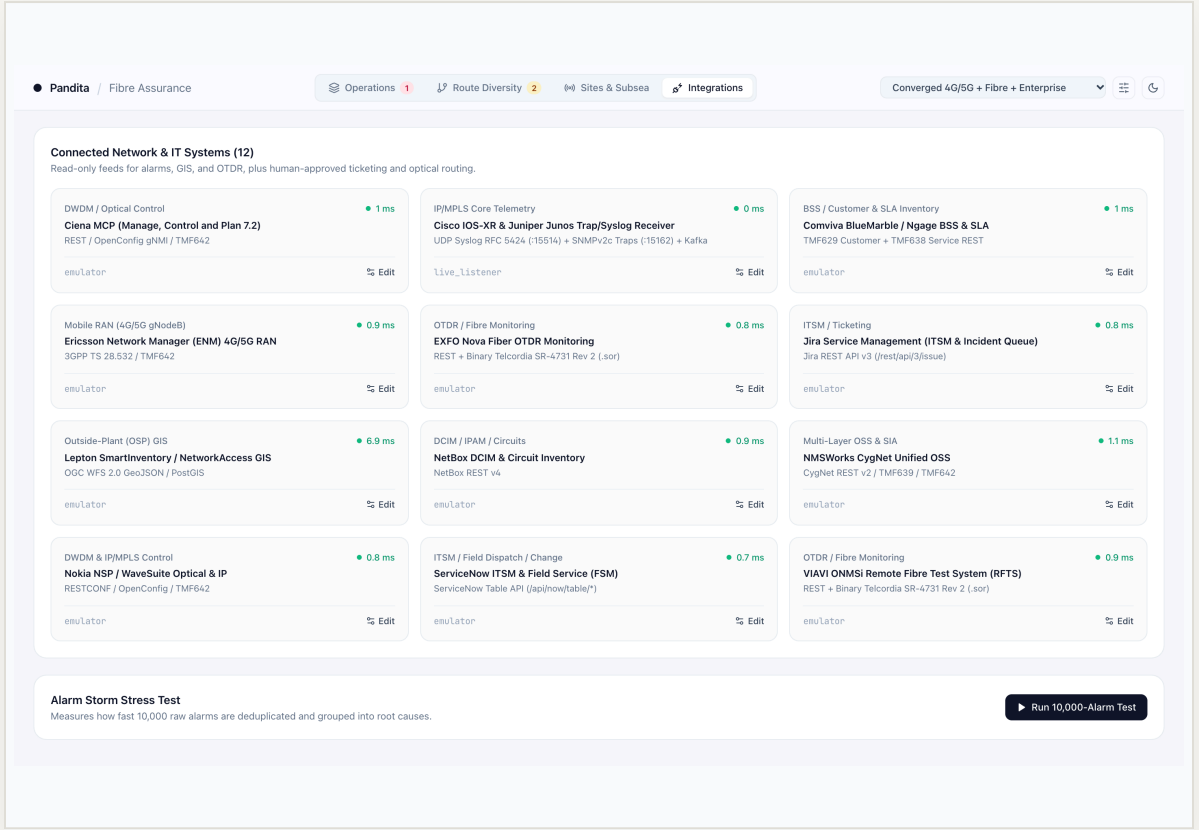
Built on Python 3.11, FastAPI, PostgreSQL 16 + PostGIS, Redpanda/Kafka and a binary Telcordia SR-4731 (.sor) codec so distances, slack offsets and disjoint routes are exact.

12 standards-native connectors

Plugs into Ciena, Nokia, VIAVI, EXFO, Ericsson, Cisco/Juniper, Lepton GIS, NetBox, ServiceNow and Jira via RESTCONF, gNMI, TMF642, WFS and Syslog/SNMP.

Cloud or air-gapped sovereign deploy

Runs in Docker or Kubernetes with zero subscriber PII, read-only southbound ingestion, human approval gates, and local LLM/embedding support for air-gapped NOCs.



Delivery approach for a four-week corridor pilot

Scope and design Week 1	Select one priority metro ring or inter-city corridor, agree clear success measures (triage speed, dispatch accuracy, quote turnaround), and connect read-only GIS and alarm feeds.
Build and integrate Weeks 2–3	Connect optical management, OTDR test feeds and ticketing workflows in shadow mode alongside your NOC, validating fault pinpointing and false-dispatch filtering on live events.
Deploy and support Week 4 onwards	Move into production with NOC team training, engineer-approved action gates, and a joint results review measuring avoided site visits, faster restoration and protected SLAs.

Your teams keep control of network changes and customer actions.

Pandita POD works with your NOC, outside-plant, IT and commercial teams through delivery.

We can start with your top three priority use cases

The first three can come from any priority, or start with a live bundle tailored to your network.

Metro Fibre & DCI

e.g., Constl

06 Data-centre interconnect, 08 Fibre-cut assurance and 10 Optical diverse-route planning.

Converged National

e.g., e&, du, Vi

08 Fibre-cut assurance, 09 Multi-layer incident resolution and 10 Optical diversity audit.

Subsea & Wholesale

e.g., PCCW Global

06 400G DCI activation, 08 Coastal vessel & fibre watch and 12 Verified RFP route sheets.